

Wannes Fraj

Robotics & Embedded Systems Engineer

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SUMMARY

Industrial Computer Engineering student at ENET'Com specializing in robotics, embedded systems, and IoT. Experienced in designing and implementing full robotic systems — from mechanical conception and embedded firmware to sensor integration and real-time control. Familiar with AI-assisted perception and autonomous navigation techniques. Seeking a summer internship in robotics or automation engineering.

EDUCATION

- **National Engineering School of Sfax (ENET'Com)** 2025 – Present
Industrial Computer Engineering Sfax, Tunisia
- **Faculty of Sciences of Monastir** 2023 – 2025
Preparatory Cycle in Physics and Chemistry Monastir, Tunisia

TECHNICAL STACK

- **Languages:** C, C++, Python, MATLAB, Dart, JavaScript
- **Robotics & Navigation:** ROS2, Nav2, SLAM, LiDAR, RViz, Gazebo, Path Planning, Sensor Fusion
- **Embedded Systems & IoT:** ESP32, Microcontrollers, MQTT, BLE, Serial Communication, Motor Control, Real-Time Firmware, PlatformIO,
- **Simulation:** Proteus 8, Gazebo
- **AI for Robotics:** Computer Vision, Obstacle Detection, LSTM-based Prediction, Scikit-learn, NumPy, Pandas

PROJECTS

- **Autonomous Heterogeneous Vehicle Navigation** (3rd Place – IEEE TSYP 2025) 2025
Tools: ROS2, Nav2, LiDAR, SLAM, Gazebo, RViz [\[in\]](#)
 - * Developed an autonomous navigation system using ROS2 Nav2 for efficient delivery tasks in dynamic environments.
 - * Implemented SLAM with LiDAR for real-time mapping, localization, and obstacle avoidance.
 - * Built and validated the full system in Gazebo simulation with RViz visualization prior to physical deployment.
 - * Awarded **3rd place** at the IEEE TSYP 2025 national competition.
- **CubeSat Thermal Prediction – Embedded AI for Space Systems** 2025
Tools: Python, LSTM, Operational Research, Synthetic Dataset [\[G\]](#)
 - * Modeled orbital dynamics and thermal behavior of a CubeSat using Python simulation and synthetic dataset generation.
 - * Trained an LSTM neural network to predict solar-facing panel orientations for real-time thermal risk management.
 - * Applied operational research algorithms for orientation classification, improving mission reliability and autonomous decision-making.
- **Smart Home – Wireless Roller Shutter Control System** 2026
Tools: ESP32, C++, MQTT, Mosquitto, BLE, Flutter, Dart, PlatformIO, Proteus 8 [\[G\]](#)
 - * Designed an IoT node for remote roller shutter control using ESP32 with Wi-Fi and Bluetooth connectivity.
 - * Developed ESP32 firmware in C++ (PlatformIO): BLE provisioning, MQTT command handling, manual override, and percentage-based positioning.
 - * Designed full circuit schematic and simulated in Proteus 8.

CERTIFICATIONS

- **Deep Learning – NVIDIA DLI** 2026

VOLUNTEERING

- **Webmaster** Jan 2026 – Present
IEEE ENET'Com WIE Student Affinity Group
 - * Designed visual content (posters, banners) using Photoshop and Premiere Pro.
 - * Coordinated event logistics and team communication for workshops and tech events.
 - * Managed online presence and publication schedules across platforms.